Integers & Algebraic Expressions

Name_____ Period____

Addition means combine.

① Draw each number. ++ ++ −−− −−−

$4 + (-6)$

② Same signs → **Party!** Add.
Different signs → **Battle!**
Cancel the pairs.

③ Who is left? ++ ++ −−− −−−

$4 + (-6) = -2$

Example 1

++++ ++++ ++

−−− −−

$-3 + 10 = 7$

Different signs:
Battle! *Cancel pairs.*

Example 2

−−−−− −−

$-5 + (-2) = -7$

Same signs:
Party! *Add.*

Subtraction means combine the opposite.

① Rewrite: add the opposite.

$5 - (-3)$
↓ ↓ flip both signs
$5 + (+3)$

② Draw each number. ③ Party! Add.

+++ ++ +++

$5 + (+3) = 8$

Example 1

$2 - 6$

++ −−−

$2 + (-6) = -4$

Example 2

$-4 - 5$

−−− −

−−− −−

$-4 + (-5) = -9$

Example 3

$-2 - (-6)$

−− +++ +++

$-2 + (+6) = 4$

Example 4

$3 - (-2)$

+++ ++

$3 + (+2) = 5$

Multiplication A(B) means A groups of B.

3(−2)
means
3 groups of −2

−− −− −−

$3(-2) = -6$

−(−2)
means
the opposite of
1 group of −2

−− → ++

$-(-2) = 2$

−2(−4)
means
the opposite of
2 groups of −4

−−−− −−−− → −8 opp. → 8

$-2(-4) = 8$

Example 1

$4(-3)$

4 groups of −3

−−− −−− −−− −−− → −12

$4(-3) = -12$

Example 2

$-(-5)$

opposite of 1 group of −5

−−−−− → −5 opp. → 5

$-(-5) = 5$

Example 3

$-5(-2)$

opposite of 5 groups of −2

−− −− −− −− −− → −10 opp. → 10

$-5(-2) = 10$

Example 4

$-3(4)$

opposite of 3 groups of 4

++++ ++++ ++++ → 12 opp. → −12

$-3(4) = -12$

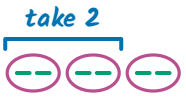
A fraction of a group

$\frac{2}{3}(-6)$ means $\frac{2}{3}$ of a group of -6

① Split -6 into 3 equal parts.

② Take 2 of the parts.

take 2

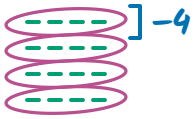


$\frac{2}{3}(-6) = -4$

Example 1

$\frac{1}{4}(-16)$

split into 4 parts, take 1

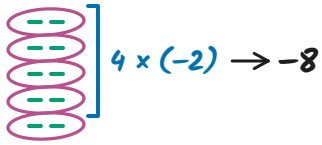


$\frac{1}{4}(-16) = -4$

Example 2

$\frac{4}{5}(-10)$

split into 5 parts, take 4



$\frac{4}{5}(-10) = -8$

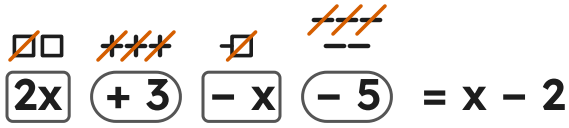
Combining Like Terms

A variable, like **x**, is an unknown number.
Draw it as a mystery box: □ = x −□ = −x

Only **like terms** combine: x's with x's, numbers with numbers.

① **Box** x terms. **Circle** numbers. Take the sign in front with them!

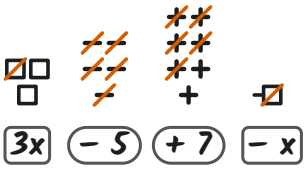
② Draw each term. ③ Cancel pairs.



$= x - 2$

Example 1

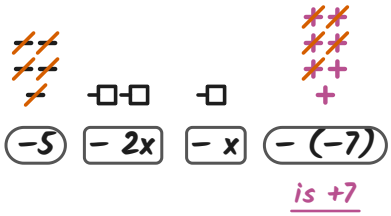
$3x - 5 + 7 - x$



$= 2x + 2$

Example 2

$-5 - 2x - x - (-7)$



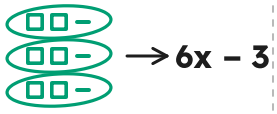
$= -3x + 2$

Distributive Property

Still **A** groups of **B**. Now **B** has 2+ terms.

$3(2x - 1)$

3 groups of $(2x - 1)$



$\rightarrow 6x - 3$

6 boxes, 3 negatives

$3(2x - 1) = 6x - 3$

$-2(x - 4)$

the opposite of 2 groups of $(x - 4)$



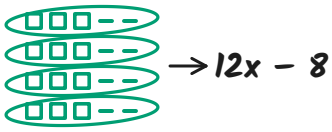
$2x - 8 \xrightarrow{\text{opp.}} -2x + 8$

$-2(x - 4) = -2x + 8$

Example 1

$4(3x - 2)$

4 groups of $3x - 2$



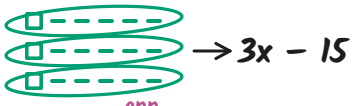
$\rightarrow 12x - 8$

$4(3x - 2) = 12x - 8$

Example 2

$-3(x - 5)$

opposite of 3 groups of $x - 5$



$\rightarrow 3x - 15$

$3x - 15 \xrightarrow{\text{opp.}} -3x + 15$

$-3(x - 5) = -3x + 15$

Distribute, then combine.

You can't combine until you know how many of each you have. Open the groups first!

$2(3x - 4) - x + 5$

① $= 6x - 8 - x + 5$ *distribute*

② $= 5x - 3$ *combine*

Example 1

$3(x + 2) - 5x$

$= 3x + 6 - 5x$

$= -2x + 6$

Example 2

$5 - 2(2x - 3)$

$= 5 + -2(2x - 3)$

$= 5 - 4x + 6$

$= -4x + 11$

subtract = add the opposite